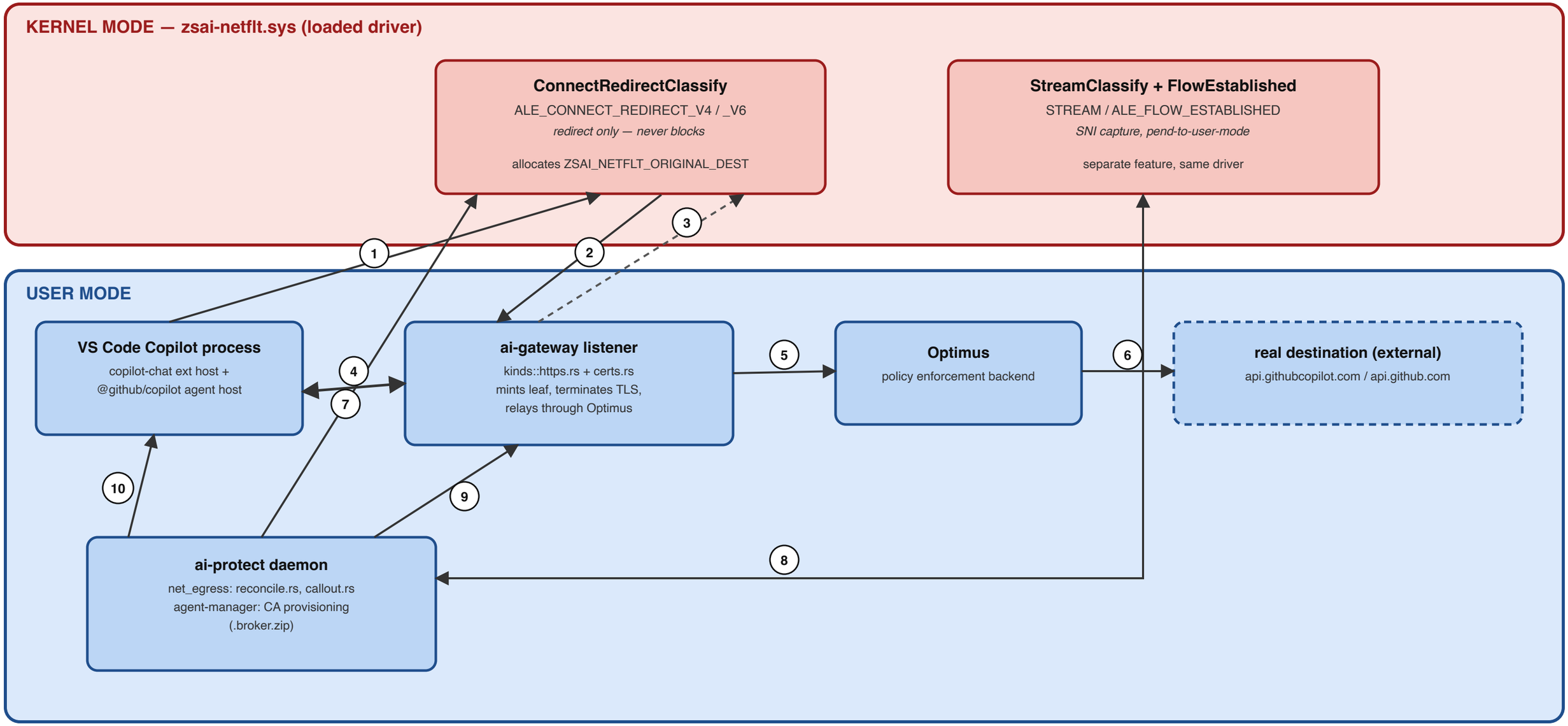
VS Code Copilot network-egress redirect — components and calls

kernel mode
user mode (everything else)



Legend

① connect() to api.githubcopilot.com:443 — client-initiated, unmodified at this point

② ConnectRedirectClassify rewrites dest to 127.0.0.1:<port>, stamps localRedirectTargetPID / Handle / Context

③ Listener recovers the original dest via SIO_QUERY_WFP_CONNECTION_REDIRECT_CONTEXT — not yet implemented

④ TLS handshake: listener mints a per-host leaf (certs.rs); client validates it against the root CA it already trusts

⑤ Listener relays decrypted, policy-visible traffic onward

⑥ Optimus forwards the request on, post-policy-decision

⑦ ai-protect daemon adds the redirect filter: FwpmFilterAdd0, filter->context = pid<<16 | port

⑧ Separate (SNI) feature, same driver: daemon ↔ StreamClassify via IOCTL_GET_FLOW / IOCTL_SET_VERDICT

⑨ ai-protect daemon delivers the CA (cert_pem + key_pem) to the listener via .broker.zip

⑩ ai-protect daemon installs the root CA into the agent's trust (NODE_EXTRA_CA_CERTS, or the OS store for vscode-copilot)

① and ⑦ are not yet exercised by a real filter; ② and ⑦'s kernel code (finding #9) is written but uncompiled/untested; ③ has no user-mode caller yet.

⑧'s SNI enforcement path is an independent, already-shipped feature sharing this driver — included here to show the driver's full kernel/user-mode split.